Project Management Principles
Revising Project Charter

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Project Management (PM) Principles

Why PM Matters

- Data engineering projects are complex, multi-disciplinary, and high-impact.
- Strong project management ensures:
  - Clear requirements and scope definition.
  - Timely delivery of reliable data pipelines.
  - Alignment with business goals.
- PM bridges the gap between data engineering teams and stakeholders.

What Makes Data Engineering Projects Different

- High data volume, velocity, and variety.
- Interdependence across data sources and systems.
- Need for scalability, quality, and governance.
- Rapidly evolving technologies (cloud, ETL tools, orchestration).
- Cross-functional collaboration between analysts, engineers, and business users.

Core Principles

1. **Clear Scope Definition** — avoid "data swamp" through strong requirements.
2. **Stakeholder Alignment** — define ownership of data products.
3. **Iterative Delivery** — deliver in manageable increments.
4. **Risk Management** — identify dependencies early (data sources, APIs).
5. **Documentation and Communication** — essential for reproducibility.

Choosing the Right Approach

Waterfall

- Best for stable data sources and well-defined pipelines.
- Predictable delivery schedule.

Agile

- Ideal for exploratory data projects or ML pipelines.
- Encourages rapid iteration and feedback.

Hybrid Approach

- Combine Waterfall structure with Agile sprints for adaptability.

Project Management Tools (*some examples*)

- **Version Control:** Git/GitHub.
- **Workflow Management:** Airflow, Dagster, Prefect.
- **Tracking:** Jira, Asana, Trello.
- **Communication:** Slack, Confluence.
- **Documentation:** Data catalogues, dbt docs.

Best Practices for Data Teams

- Build modular and reusable pipeline components.
- Maintain strong versioning and testing discipline.
- Automate quality checks and data validation.
- Track data lineage and monitor SLAs.
- Encourage cross-team knowledge sharing.

Typical Pitfalls and How to Avoid Them

- **Scope Creep:** Use clear acceptance criteria.
- **Data Quality Issues:** Automate validation checks.
- **Communication Gaps:** Maintain consistent stakeholder updates.
- **Underestimating Complexity:** Include buffer time for testing.
- **Tool Overload:** Choose tools that integrate well with your ecosystem.

- Effective project management ensures scalable and trustworthy data systems.
- Collaboration and transparency are key to long-term success.
- Align engineering efforts with business value delivery.

"Good data engineering starts with good project management."

Revising Project Charter

1. Purpose and Justification

Key Questions:

- What problem or opportunity does this project address?
- Why is this project important to the organization?
- What business goals or strategic objectives does it support?
- What are the expected benefits or outcomes?

Response:

2. Objectives and Scope

Key Questions:

- What are the specific goals or deliverables?
- What is included in the project scope (and what is out of scope)?
- How will success be measured? ($\longleftarrow$ *very very important!*)

Response:

3. Stakeholders and Roles

Key Questions:

- Who is the project sponsor and project manager?
- Who are the key stakeholders and their responsibilities?
- Who will be impacted by the project?

Response:

4. Timeline and Milestones

Key Questions:

- What is the estimated project duration?
- What are the major milestones or phases?
- Are there critical deadlines or dependencies?

Response:

5. Resources and Budget

Key Questions:

- What resources (people, tools, facilities) are needed?
- What is the estimated cost or funding source?
- How will the budget be managed?

Response:

6. Risks, Assumptions, and Constraints

Key Questions:

- What are the main risks and potential obstacles?
- What assumptions are being made?
- What constraints must the project operate under?

Response:

7. Governance and Approval

Key Questions:

- How will decisions be made and communicated?
- What is the escalation process for issues?
- Who has authority to approve changes or decisions?
- Who must approve this charter to launch the project?

Response: